

Detection and Classification of Diabetic Retinopathy Using EfficientNetB0: A Lightweight Model With Transfer Learning Approach

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Introduction

- Diabetic retinopathy (DR) is a gradual consequence of diabetes that destroys the retinal blood vessels and can lead to irreversible blindness.
- There are 5 stages of DR:

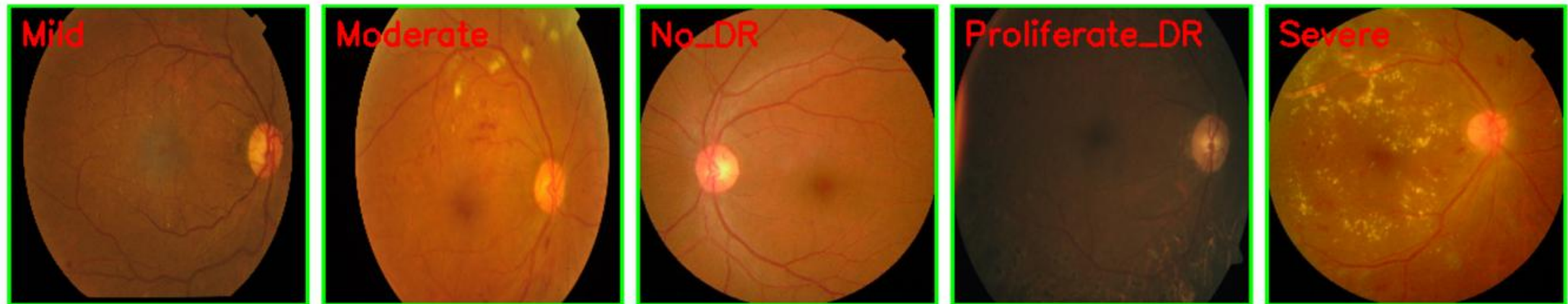


Fig.1: Retinal Images of 5 stages of Diabetic Retinopathy

Introduction

Stages	Description
No DR	No detectable signs of retinopathy are present.
Mild	Small bulges form in the blood vessels in the retina, and they may begin to leak.
Moderate	Blood vessels become more distorted and swollen, and blood flow is further restricted.
Proliferative DR	The most advanced stage, characterized by the growth of new, abnormal blood vessels in the retina. These fragile vessels can leak blood into the vitreous, clouding vision and potentially leading to severe vision impairment or blindness.
Severe	Blockages in the blood vessels deprive parts of the retina of oxygen. This stage can lead to new, fragile blood vessels beginning to grow.

Table 1: Description of the 5 Stages of DR

Problem Statement

- DR is the primary cause of vision loss in middle-aged adults in many areas, and the fifth major preventable cause of blindness worldwide in 2021.[2]
- Research shows that DR poses a significant public health challenge, affecting approximately one in four diabetic patients worldwide.[4]
- Approximately 10–11% of diabetics with DR progress to vision-threatening stages (severe DR or PDR) annually. [4]
- Though the early detection of DR is critical for the prevention of blindness, these stages usually do not show any symptoms and may go unnoticed until irreversible retinal damage occurs. [4]
- Regular retinal checkups enable experts to detect subtle retinal changes before patients experience vision loss.[1]

Contribution

Many patients lack access to routine screenings, leading to delayed diagnosis and preventable vision loss [4]. To address this gap, we propose an automated deep learning system with three key advances:

1. An EfficientNetB0-based model for the DR stage classification, optimized for early detection with high accuracy and low computational cost.
2. A strategy to resolve class imbalance of the APTOS dataset using augmentation and improved feature extraction through erosion and histogram equalization.
3. A comprehensive comparison of the proposed model with other state-of-the-art models, demonstrating superior performance with an accuracy of 96.00%.

Literature Review

Ref.	Authors	Model Used	Accuracy
[5]	Nandhini et al. [2023]	DenseNet-169, DiaNet	Accuracy 90.02% (DiaNet)
[6]	Mahendar et al. [2023]	Customized CNN	Accuracy 91%
[7]	Duvvuri et al. [2023]	Customized CNN With Image processing	Accuracy after Preprocessing 86%
[8]	Baskar et al. [2024]	AlexNet and DenseNet-169 with transfer learning	Best F1 score achieved at PDR class: 69%
[9]	Dasari et al. [2023]	ResNet50 with Transfer Learning	Accuracy 90%
[10]	Priya et al. [2023]	VGG-16, MobileNetV2	Accuracy 92% (MobileNetV2)
[11]	Kumar et al. [2024]	RetinaRes	Accuracy 92%
[12]	Dasari et al. [2024]	Capsule Network	Accuracy 95.41%

Dataset

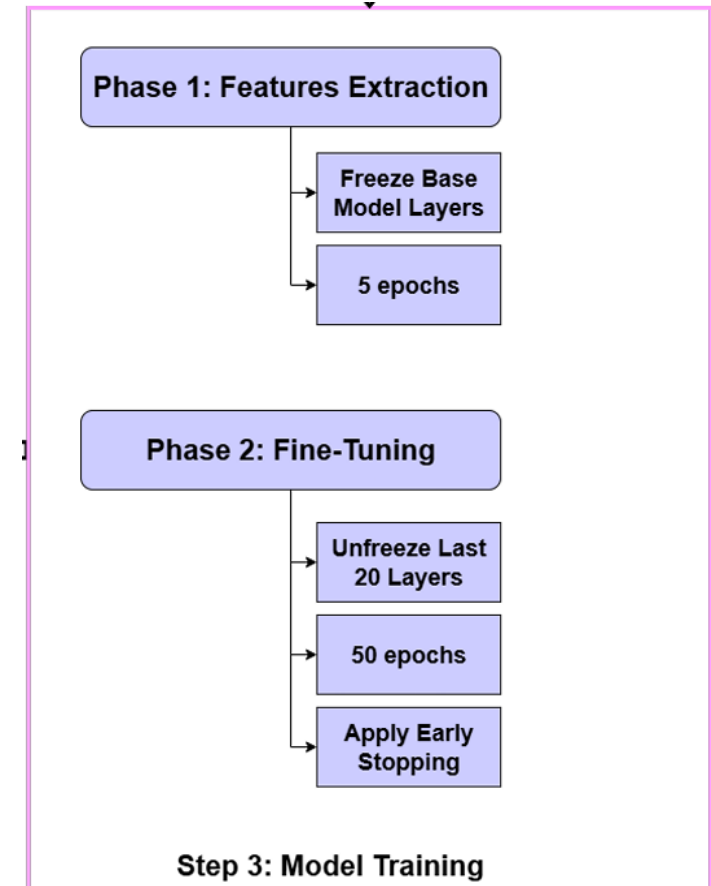
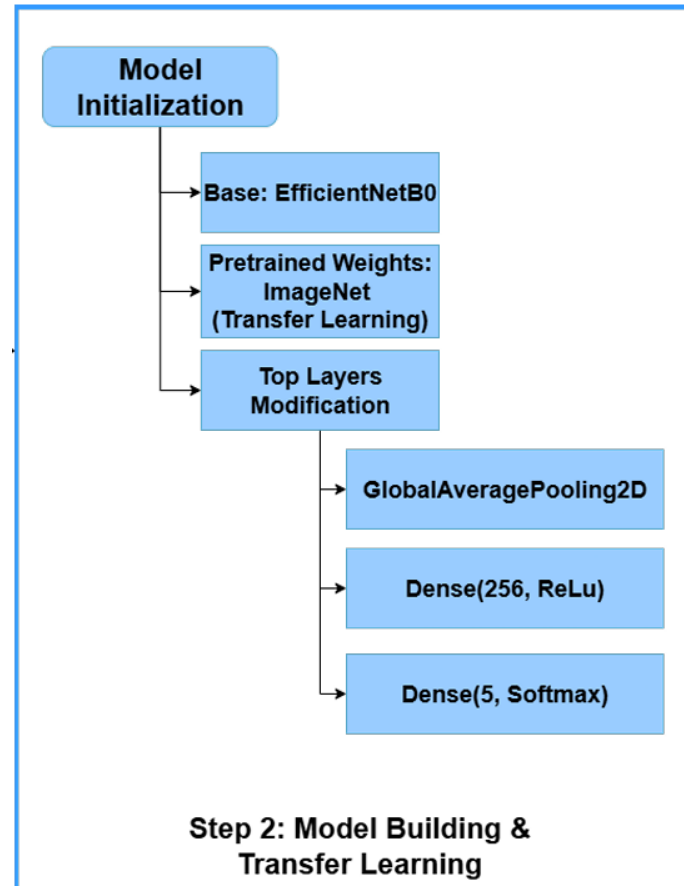
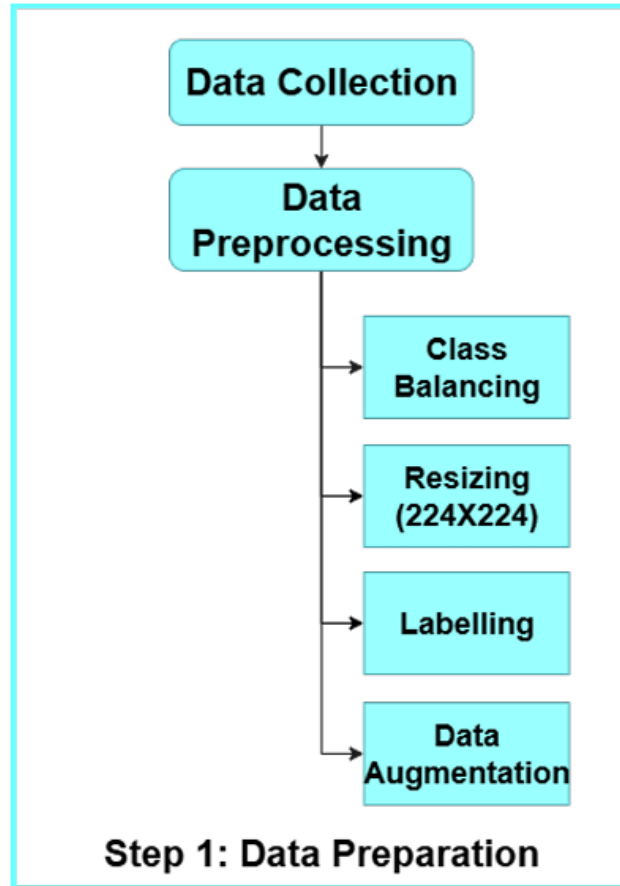
- APTOS-2019 (Asia Pacific Tele-Ophthalmology Society)
- EyePACS (Eye Picture Archive Communication System)

(Both datasets contain high-quality fundus images)

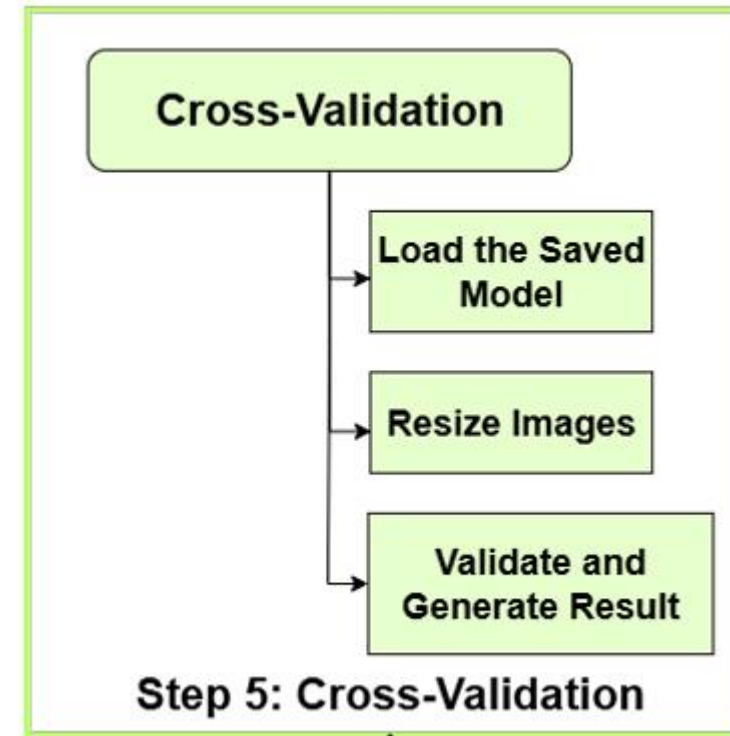
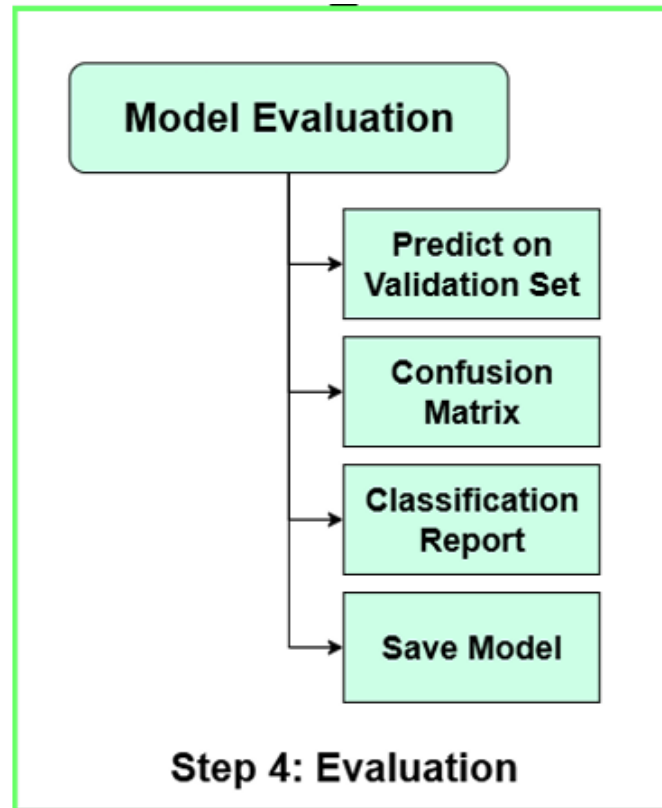
Category	APTOS	EyePACS
No DR	1805	25802
Mild	370	2432
Moderate	999	5288
Proliferative DR	295	872
Severe	193	708

Table 2: Image count of the five classes of both datasets.

Methodology



Methodology



Model Evaluation

Accuracy

Precision

Recall

F1 Score

**Classification
Report**

**Accuracy-Loss
Curve**

Cross Validation

Category	Before Augmentation	After Augmentation
No DR	1805	1000
Mild	370	1000
Moderate	999	1000
Severe	193	1000
Proliferate DR	295	1000

Table 3: Image count of the five classes.

Augmentation	Accuracy	Precision	Recall	F1-Score
After	96.00	96.09	96.00	95.98
Before	81.37	80.78	81.37	80.72

Table 4: Scores of the Evaluation Matrix of our Proposed model.

Category	Precision	Recall	F1-Score	Support
Mild	.93	.99	.99	200
Moderate	0.98	0.90	0.94	200
No_DR	0.98	0.96	0.97	200
Proliferate_DR	0.96	0.94	0.95	200
Severe	0.96	1.00	0.98	200
accuracy			0.96	1000
mircro avg	0.96	0.96	0.96	1000
weighted avg	0.96	0.96	0.96	1000

Table 5: Classification Report of our Proposed Model

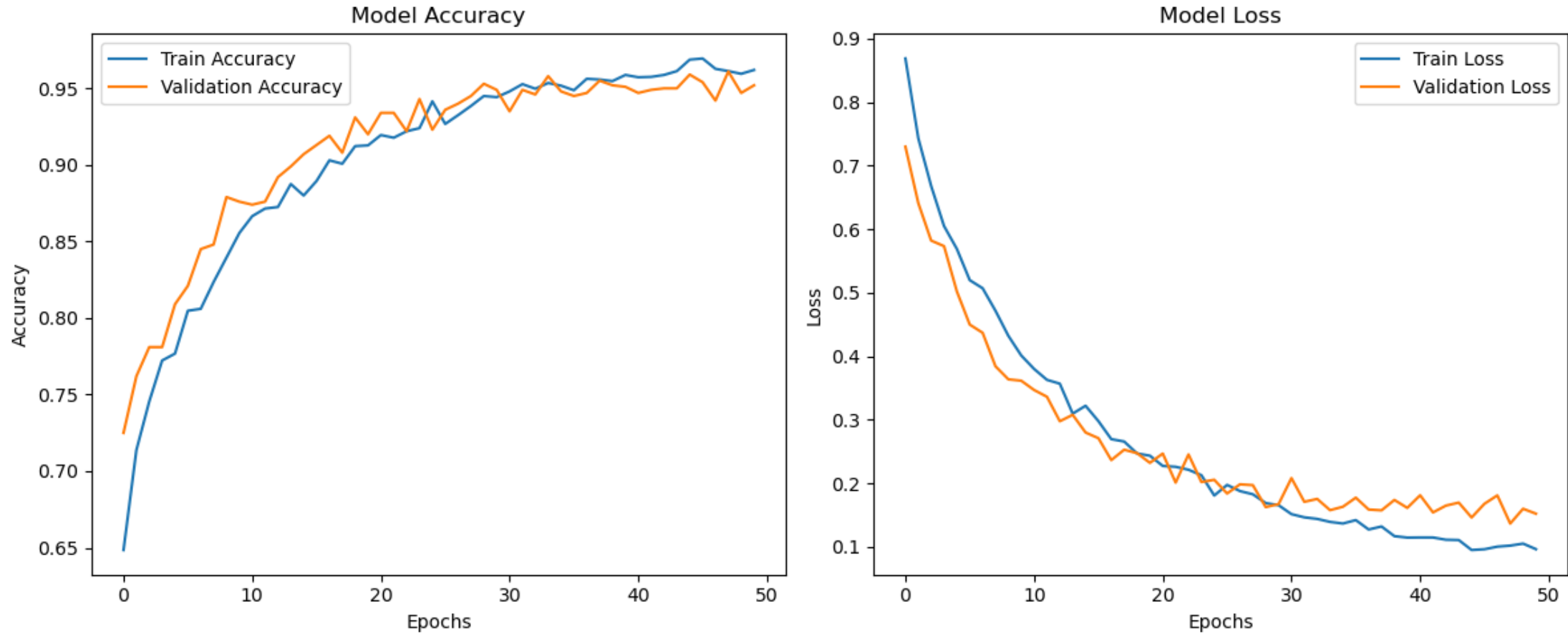


Fig.2: Accuracy and Loss curve.

Cross Validation

Augmentation	Accuracy	Precision	Recall	F1-Score
After	86.10%	86.79%	86.10%	86.05%

Table 4: Scores of the Evaluation Matrix of our Proposed model on the Validation Dataset.

Performance Comparison

Works	Model Used	Accuracy
Priya et al. [10]	VGG-16, MobileNetV2	92% (MobileNetV2)
Kumar et al. [11]	RetinaRes	92%
Nandhini et al. [5]	DenseNet-169	90.02%
Dasari et al. [12]	Capsule Network	95.41%
Our Proposed	EfficientNetB0	96.00%

Table 5: Comparison with other works.

Limitations & Future Work

- **Data Augmentation:** Helped address class imbalance, but may not fully reflect real-world variations.
- **Model Evaluation:** Performed well on test data, but not yet validated in real clinical settings.
- **Future Data Collection:** Real fundus images from diverse patient populations are needed to handle class imbalance without relying on augmentation.
- **Clinical Integration & XAI:** Embedding the model into clinical workflows is essential for reliability; using Grad-CAM can enhance interpretability and build clinician trust.

Conclusion

- This study validates a customized EfficientNetB0 model for diabetic retinopathy classification, achieving 96.0% accuracy on the APTOS dataset.
- Through transfer learning, data augmentation, and preprocessing, the model does very well in detecting severe stages but faces challenges with moderate DR due to subtle retinal variations.
- It outperformed DenseNet-169, ResNet50, and Capsule Network in both accuracy and computational efficiency, making it suitable for real-time clinical use.

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